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CHM 101: General Chemistry I - Atoms, Bonding and Reactions

The structure-and-bonding foundations behind every chemistry course you will take

FACULTY PRIMER

Faculty Primer

Faculty of Physical Sciences
Pure Chemistry | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Why this pack exists

CHM 101 rewards students who can move between the microscopic picture and the calculation. This pack connects atomic structure, periodicity, bonding and stoichiometry, then shows how to set up mole calculations that do not collapse under exam pressure.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Describe atomic structure, electronic configuration and periodic trends with correct reasoning.
- Predict bond type and molecular shape using electronegativity and VSEPR theory.
- Balance equations and solve mole, limiting-reagent and concentration problems.
- Explain the behaviour of acids, bases and buffers at an introductory level.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Atomic structure and periodicity

The periodic table is not a chart to memorise; it is a map of electron configuration.

- Atomic number is the number of protons; mass number is protons plus neutrons.
- Isotopes have the same atomic number but different mass numbers.
- Electrons fill orbitals in order of increasing energy, following the Aufbau principle, Pauli exclusion principle and Hund's rule.
- Atomic radius decreases across a period and increases down a group.
- Ionisation energy increases across a period and decreases down a group.

- Electron affinity and electronegativity follow similar trends to ionisation energy.

> **Explain, do not just state**
> A question asking why ionisation energy increases across a period wants the effective nuclear charge argument, not a restatement of the trend.

Bonding and molecular shape

Bonding is about electrons: transfer creates ions, sharing creates covalent bonds, and unequal sharing creates polarity.

- Ionic bonding: metal to non-metal, lattice structures, high melting points.
- Covalent bonding: shared electron pairs, discrete molecules or giant networks.
- Metallic bonding: delocalised electrons explain conductivity and malleability.
- Intermolecular forces: London dispersion, dipole-dipole and hydrogen bonding, in increasing strength.
- VSEPR: electron pairs repel and adopt the arrangement that maximises separation.
- Polar bonds do not guarantee a polar molecule; symmetry can cancel the dipoles.

Moles and stoichiometry

Almost every calculation in CHM 101 starts with converting something into moles.

1. Write and balance the equation first.
2. Convert given mass, volume or concentration into moles.
3. Use the balanced equation ratio to find the moles of the required substance.
4. Identify the limiting reagent when two reactant amounts are given.
5. Convert moles back to the requested unit: mass, volume of gas at stated conditions, or concentration.
6. Check significant figures and units.

> **The limiting reagent test**
> Divide the moles of each reactant by its coefficient. The smallest quotient identifies the limiting reagent - not the smallest mass.

Atoms, moles and stoichiometry

Chemistry is quantitative. The mole is the bridge between the microscopic and the measurable: one mole contains Avogadro's number of particles, and its mass in grams equals the relative molecular mass. Almost every calculation in first-year chemistry runs through the mole.

Stoichiometry uses the balanced equation as a ratio machine. Balance first, convert masses to moles, apply the ratio from the coefficients, convert back to the requested quantity. The limiting reagent is the one that runs out first, and it determines the maximum yield; percentage yield compares what you got with that maximum.

The empirical formula comes from percentage composition, the molecular formula from the molar mass. These are routine calculations, and marks are lost mainly through unit errors and unrounded intermediate values.

Bonding and structure

Ionic bonding transfers electrons between atoms with large electronegativity difference, producing a lattice held by electrostatic attraction. Covalent bonding shares electrons, producing discrete molecules or giant structures. Metallic bonding delocalises electrons across a lattice, which explains conductivity and malleability.

Intermolecular forces - London dispersion, dipole-dipole and hydrogen bonding - determine boiling point, solubility and viscosity. This is why water, with hydrogen bonding, boils far above hydrogen sulphide despite a similar shape, and why ice floats.

VSEPR theory predicts molecular shape from electron-pair repulsion: pairs arrange to minimise repulsion, lone pairs repel more strongly than bonding pairs. Shape determines polarity, and polarity determines physical behaviour and reactivity.

> **Balance before you calculate**
> An unbalanced equation will produce a confidently wrong answer. Check atom counts, then charges.

Equilibria, acids and redox

Reversible reactions reach dynamic equilibrium where forward and reverse rates are equal. Le Chatelier's principle predicts the direction of response to concentration, pressure and temperature changes; it does not predict the rate, and it does not tell you how fast equilibrium is reached.

Acids and bases are described by Brønsted-Lowry proton transfer and by Lewis electron-pair donation. pH is the negative logarithm of hydrogen ion concentration, and buffer solutions resist pH change because they contain a conjugate pair in comparable amounts.

Redox reactions involve changes in oxidation state. Assigning oxidation numbers correctly is the key skill; balancing redox equations in acidic or basic solution requires adding water and hydrogen ions in the correct order.

WORKED EXAMPLES

Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

1. A limiting reagent calculation

You react 4.6 g of sodium with 8.0 g of chlorine. Which is limiting, and what mass of sodium chloride forms?

1. Write and balance: 2Na + Cl2 -> 2NaCl.
2. Convert to moles: Na = 4.6 / 23 = 0.20 mol; Cl2 = 8.0 / 71 = 0.113 mol.
3. Apply the ratio: 2 mol Na needs 1 mol Cl2, so 0.20 mol Na needs 0.10 mol Cl2.
4. Compare with available: 0.113 mol Cl2 is available, so Cl2 is in excess and Na is limiting.
5. Moles of NaCl from Na: 0.20 mol.
6. Mass of NaCl: 0.20 x 58.5 = 11.7 g.

> **What to carry forward**
> Convert to moles, apply the coefficient ratio, compare, then convert back.

2. Predicting a shape and polarity

Predict the shape of ammonia and whether it is polar.

1. Count valence electrons: N (5) + 3 H (3) = 8.
2. Draw the Lewis structure: three N-H bonds plus one lone pair on N.
3. Apply VSEPR: four electron pairs, one lone pair, hence trigonal pyramidal.
4. Assess symmetry: the lone pair makes the arrangement asymmetric.
5. Conclude: bond dipoles do not cancel, so ammonia is polar.
6. Relate to behaviour: high solubility in water and hydrogen bonding.

> **What to carry forward**
> Electrons, then shape, then symmetry, then polarity. Each step uses the previous one.

EXAM TECHNIQUE

Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

Q1. Explain the difference between ionic and covalent bonding and how the difference affects physical properties.

15 marks

1. Define each in terms of electron transfer and sharing.
2. Relate electronegativity difference to bond type.
3. Compare melting points, conductivity and solubility.
4. Use a specific example of each.

Q2. State Le Chatelier's principle and apply it to the Haber process.

15 marks

1. State the principle.
2. Write the equation and note the exothermic direction.
3. Predict the effect of pressure, temperature and removal of ammonia.
4. Explain the industrial compromise between yield and rate.

> **Where marks are lost**
> Confusing equilibrium yield with reaction rate.

REFERENCE

Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Mole. The amount containing Avogadro's number of particles.

Molar mass. The mass of one mole of a substance in grams.

Limiting reagent. The reactant that is exhausted first and so limits product formation.

Empirical formula. The simplest whole-number ratio of atoms in a compound.

Electronegativity. An atom's tendency to attract shared electrons.

Hydrogen bond. A strong intermolecular attraction between a hydrogen bonded to N, O or F and a lone pair.

VSEPR. A theory predicting molecular shape from electron-pair repulsion.

Dynamic equilibrium. A state where forward and reverse reaction rates are equal.

Buffer. A solution that resists pH change when acid or base is added.

Oxidation number. A bookkeeping charge assigned to an atom in a compound.

Stoichiometry. Quantitative relationships between reactants and products in a reaction.

Percentage yield. Actual yield expressed as a percentage of theoretical yield.

Your four-week plan

Week 1 - Moles

Quantities.

- Convert between mass, moles and particles for ten substances.
- Determine empirical formulas from percentages.
- Solve three limiting reagent problems.

Week 2 - Bonding

Structure.

- Draw Lewis structures for ten molecules.
- Predict shapes with VSEPR.
- Compare boiling points using intermolecular forces.

Week 3 - Equilibrium

Reversibility.

- Write equilibrium expressions.
- Apply Le Chatelier to three systems.
- Calculate pH for strong and weak acids.

Week 4 - Redox

Electron transfer.

- Assign oxidation numbers in ten compounds.
- Balance three redox equations.
- Identify oxidising and reducing agents.

Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Any first-year general chemistry text covering stoichiometry, bonding and equilibrium.
- A calculations workbook for mole and titration problems.
- A laboratory manual section on safe handling of acids and oxidising agents.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. How many moles are in 18 g of water?

Answer: Molar mass of water is 18 g/mol, so 18 g is 1 mole.

2. Why is the first ionisation energy of sodium lower than that of magnesium?

Answer: Sodium's outer electron is in the 3s orbital and removing it gives a stable noble-gas configuration; magnesium's electron is held by a greater effective nuclear charge, so more energy is needed.

3. Predict the shape of methane and explain it.

Answer: Tetrahedral, because four bonding pairs around carbon repel equally and adopt the arrangement that maximises separation with 109.5 degree angles.

4. 20 g of calcium carbonate reacts with excess acid. Find the moles of carbon dioxide produced.

Answer: Molar mass of CaCO3 is about 100 g/mol, so 20 g is 0.2 mol. The reaction is 1:1, so 0.2 mol of CO2 is produced.

5. Is carbon dioxide polar?

Answer: No. The C=O bonds are polar but the linear geometry makes the dipoles cancel.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Write electron configurations for the first 20 elements from memory.
- [ ] Sketch the periodic trends for radius, ionisation energy and electronegativity with reasons.
- [ ] Draw VSEPR shapes for five common molecules and label bond angles.
- [ ] Solve ten stoichiometry problems including three limiting-reagent cases.
- [ ] Make a one-page summary distinguishing ionic, covalent and metallic bonding.
- [ ] Perform a titration calculation from raw readings without a template.
- [ ] Attempt one past CHM 101 objective paper and review every wrong answer.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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